# Failure budgets and service objectives

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

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## Prepare the purpose, prerequisites and materials

> Prepare Failure budgets and service objectives: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Evaluation pipelines

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of evaluation pipelines

Reference example, not the only acceptable wording: Teacher explanation: Evaluation pipelines preserve case versions, procedure versions and raw outcomes before producing metrics. A rerun must identify what changed.

Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

### Testing model-dependent behaviour

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of testing model-dependent behaviour

Reference example, not the only acceptable wording: Teacher explanation: Model-dependent behaviour may vary between runs.

Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence. Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions.

Repeats estimate observed variation for this case; they are not additional independent users.

### Authentication and authorisation

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of authentication and authorisation

Reference example, not the only acceptable wording: Teacher explanation: Authentication identifies an actor; authorisation decides whether that actor may perform a particular operation on a particular resource.

Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Two 45-minute sessions. **Service objective**s define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

### Prerequisite answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

### End goal for the pictorial case

Explain **failure budget**s and **service objective**s using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain and model the mechanism

> Model Failure budgets and service objectives using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

**Service objective**s define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

### Explain Failure budgets and service objectives · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For a fictional objective at least 98% of 200 **eligible request**s must pass all contract checks. Allowed failures are 2%*200=4. Count every eligible failure once; report excluded requests separately.

## Explain Failure budgets and service objectives · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

| Term | Meaning |
| --- | --- |
| Service objective | A declared measured target over a specified scope |
| Failure budget | The failures allowed under that target |
| Eligible request | A request included under the declared denominator rule |

### Failure budgets and service objectives — evidence trace

**INPUT AND SCOPE**

In the interval 200 requests are eligible and 5 fail.

**CHECK OR CALCULATION**

For a fictional objective at least 98% of 200 eligible requests must pass all contract checks.

**CONCLUSION AND LIMIT**

195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

## Explain Failure budgets and service objectives · 3

Guided: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only. Qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

In the interval 200 requests are eligible and 5 fail.

195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

## Model the reasoning aloud · 4

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

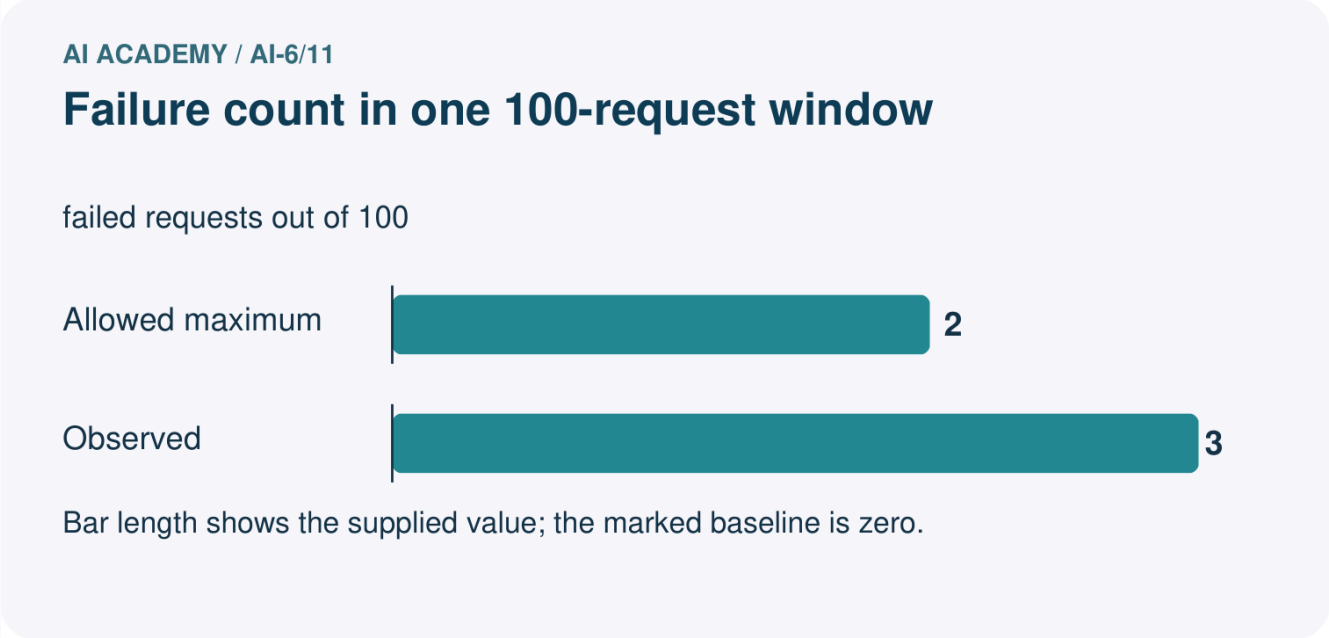
### A pictorial worked case: Failure budgets and service objectives

A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.

### The facts we will use

#### Supplied facts

Current window has 97 successes and 3 failures. Success includes a correct refusal within the agreed time limit.




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/11

# Failure budgets and service objectives

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Define success

Use the same predeclared correctness and timing criteria.



## 2. Compute allowance

The target permits at most 2 failures in 100 requests.



## 3. Compare record

Three failures misses the target by one request.

**CHANGE ONE THING**

Repair one failed request into a success in this window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Define success | Use the same predeclared correctness and timing criteria. |
| 2. Compute allowance | The target permits at most 2 failures in 100 requests. |
| 3. Compare record | Three failures misses the target by one request. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

### What this example does not establish

#### Keep the stated limit

Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

### Change one thing in the pictured case

#### Change one thing

Change: Repair one failed request into a success in this window.

#### Predicted result

**Prediction**: 98 successes and 2 failures meet the stated window target.

#### Why the result follows

Reason: One recorded outcome changes while the denominator stays 100.

### Ask while showing the picture

#### Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Failure budgets and service objectives, reduce support deliberately and check individual reasoning before discussing answers.

| Time or stage | Teacher action |
| --- | --- |
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation. |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

### Reduce help deliberately

#### Completion practice

Completion task: In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

### Reasoning to listen for

Teacher check: Teacher explanation: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Failure budget**s and **service objective**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Expected repair

Expected correction: Meeting one interval objective does not establish every future interval or zero critical failures.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

**Reasoned teacher answer**

Use the same predeclared correctness and timing criteria. The target permits at most 2 failures in 100 requests. Three failures misses the target by one request.

Changed case: 98 successes and 2 failures meet the stated window target. One recorded outcome changes while the denominator stays 100.

Changed-case reasoning: 98 successes and 2 failures meet the stated window target. One recorded outcome changes while the denominator stays 100.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the investigation and diagnose misconceptions

> Run the Failure budgets and service objectives investigation and use the first unsupported step to choose a teaching response.

The worked result establishes every future case: Meeting one interval objective does not establish every future interval or zero critical failures.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Failure budgets and service objectives questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Guided and exit questions

Guided: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only. Qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### W1 Explain the mechanism

#### Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

#### Teacher reasoning · W1

Teacher explanation: **Service objective**s define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

For a fictional objective at least 98% of 200 **eligible request**s must pass all contract checks. Allowed failures are 2%*200=4. Count every eligible failure once; report excluded requests separately.

### W2 Trace the worked case

#### Workbook question · W2

In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

#### Teacher reasoning · W2

Teacher explanation: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

### W3 Audit the limit

#### Workbook question · W3

A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

#### Teacher reasoning · W3

Teacher explanation: Meeting one interval objective does not establish every future interval or zero critical failures.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

### W4 Apply a new case independently

#### Workbook question · W4

Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

#### Teacher reasoning · W4

Teacher explanation: Allowed four; 75/80=93.75%, so objective fails. Unknown eligibility cannot be silently excluded.

### W5 Design a revealing follow-up

#### Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

#### Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Meeting one interval objective does not establish every future interval or zero critical failures.

The new case should test this particular **mechanism** rather than merely restate the original result.

### Independent assessment and fresh retry

#### Teacher key · Independent assessment and fresh retry

W4: Allowed four; 75/80=93.75%, so objective fails. Unknown eligibility cannot be silently excluded. Require intermediate reasoning and the stated scope. Fresh retry: Objective 90% over 50 requests with four failures.

Expected reasoning: Allowed five; success 46/50=92%, so interval meets objective. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: Allowed four; 75/80=93.75%, so objective fails. Unknown eligibility cannot be silently excluded.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect the project, support and next lesson

> Connect Failure budgets and service objectives to Grounded Learning Service, adapt support and plan a later changed-case check.

Add a labelled **failure budget**s and **service objective**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **failure budget**s and **service objective**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Logs traces and privacy. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

### Plan the delayed return

Suggested return points in this level: ai-6/12, ai-6/14, ai-6/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Failure budgets and service objectives lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

### Supported practice answer

Reasoned practice answer: Teacher explanation: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

Check every learner before discussion: Use the worked case for **Failure budget**s and **service objective**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Meeting one interval objective does not establish every future interval or zero critical failures.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Repair one failed request into a success in this window. Require a **prediction** and a reason before comparing. Expected result: 98 successes and 2 failures meet the stated window target.

Reason: One recorded outcome changes while the denominator stays 100.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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TEACHER GUIDE · 8 OF 9

## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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## Visual concept journey

> Show one Failure budgets and service objectives graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Failure budgets and service objectives: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s. Supplied facts: Current window has 97 successes and 3 failures.

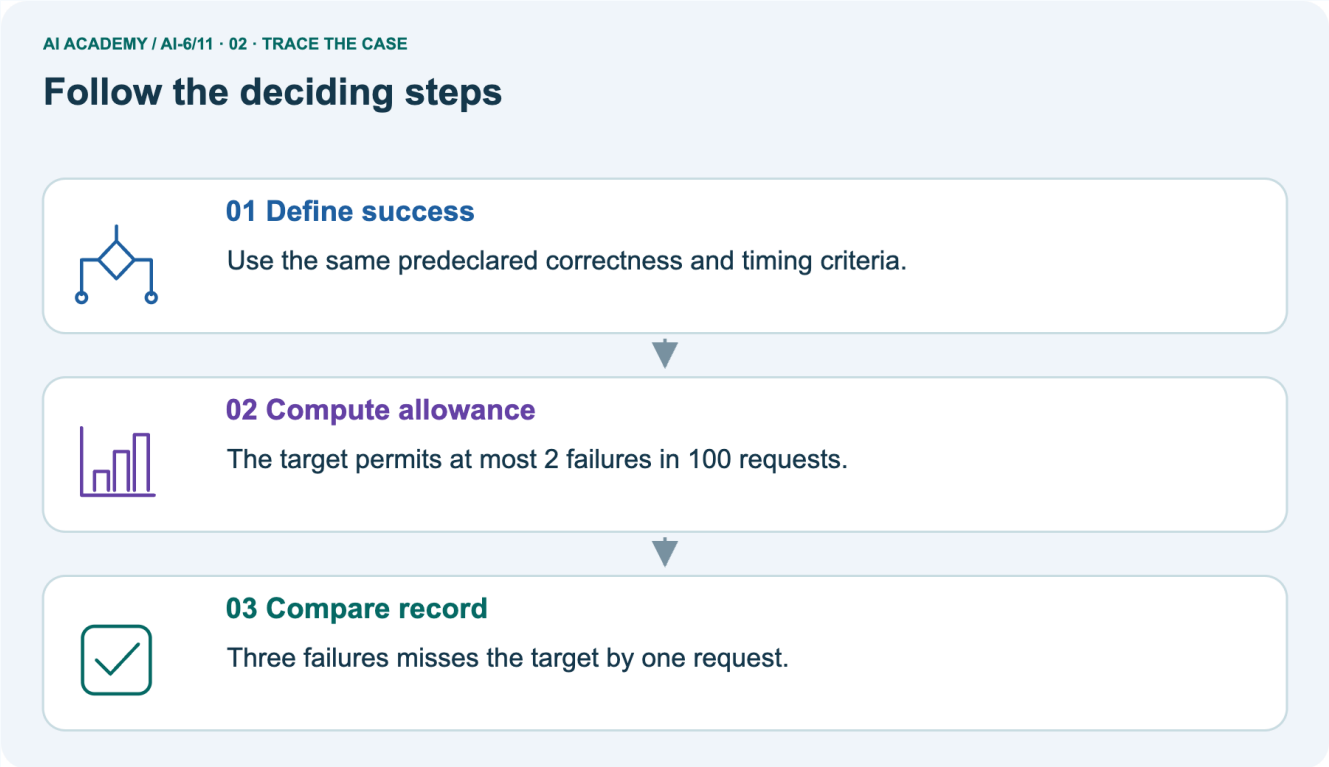
Success includes a correct refusal within the agreed time limit.

> Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Define success: Use the same predeclared correctness and timing criteria. → Compute allowance: The target permits at most 2 failures in 100 requests. → Compare record: Three failures misses the target by one request.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Failure count in one 100-request window




Follow the labelled picture with the matching explanation. Open the vector diagram

Failure count in one 100-request window. Units: failed requests out of 100. Allowed maximum: 2. Observed: 3.

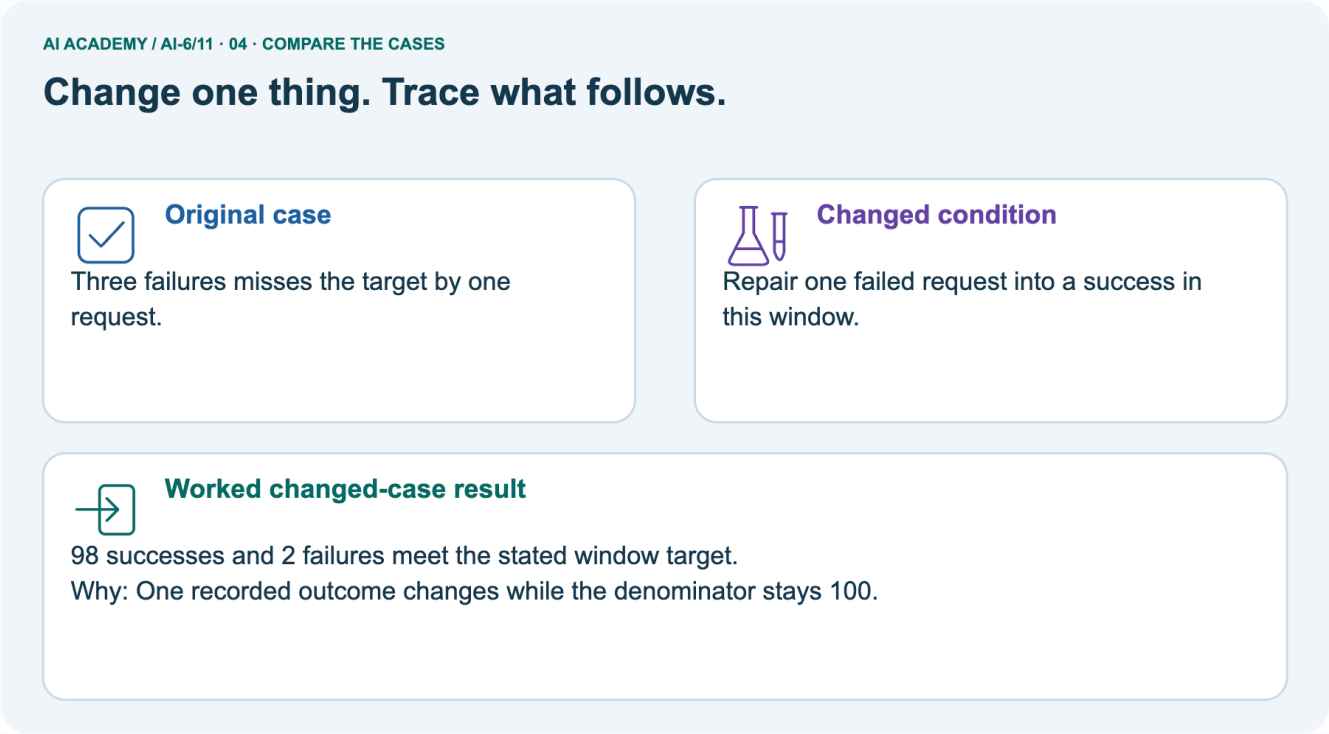
Read the labels and values. Explain the relationship the model shows.

#### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three failures misses the target by one request. Changed condition: Repair one failed request into a success in this window. Expected result: 98 successes and 2 failures meet the stated window target.

Reason: One recorded outcome changes while the denominator stays 100.

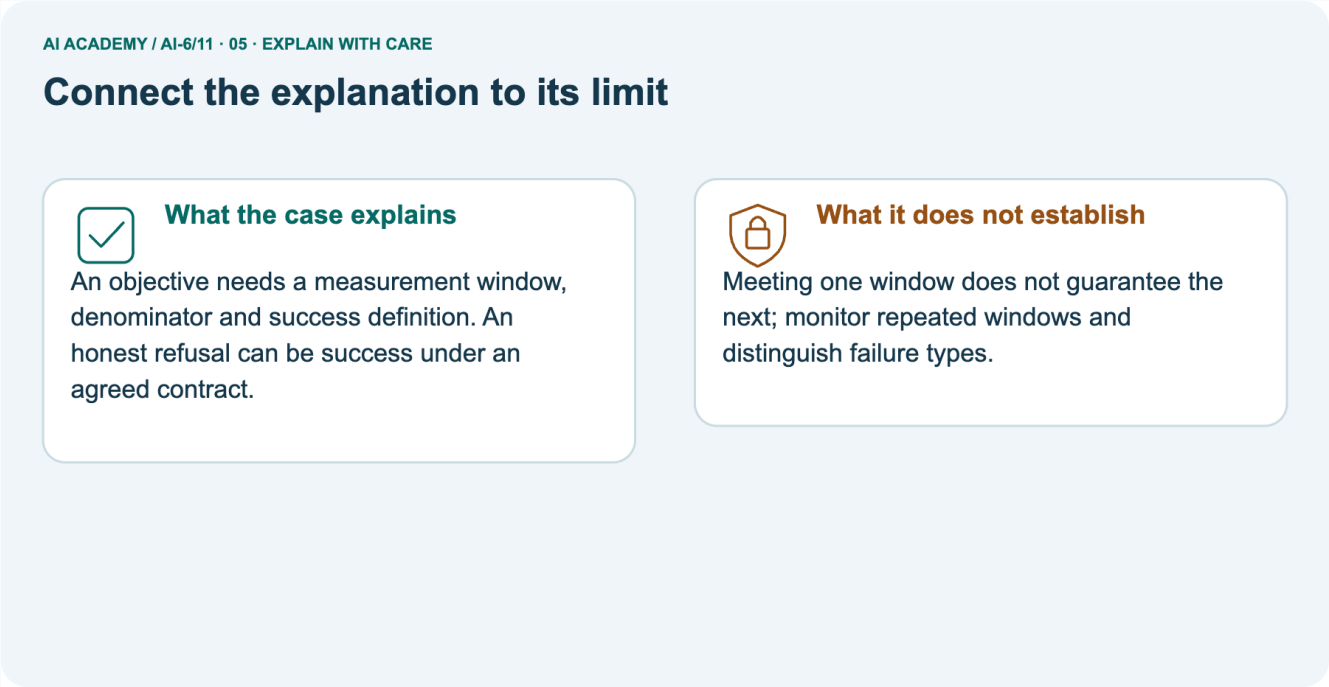
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: One recorded outcome changes while the denominator stays 100.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

Limit: Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

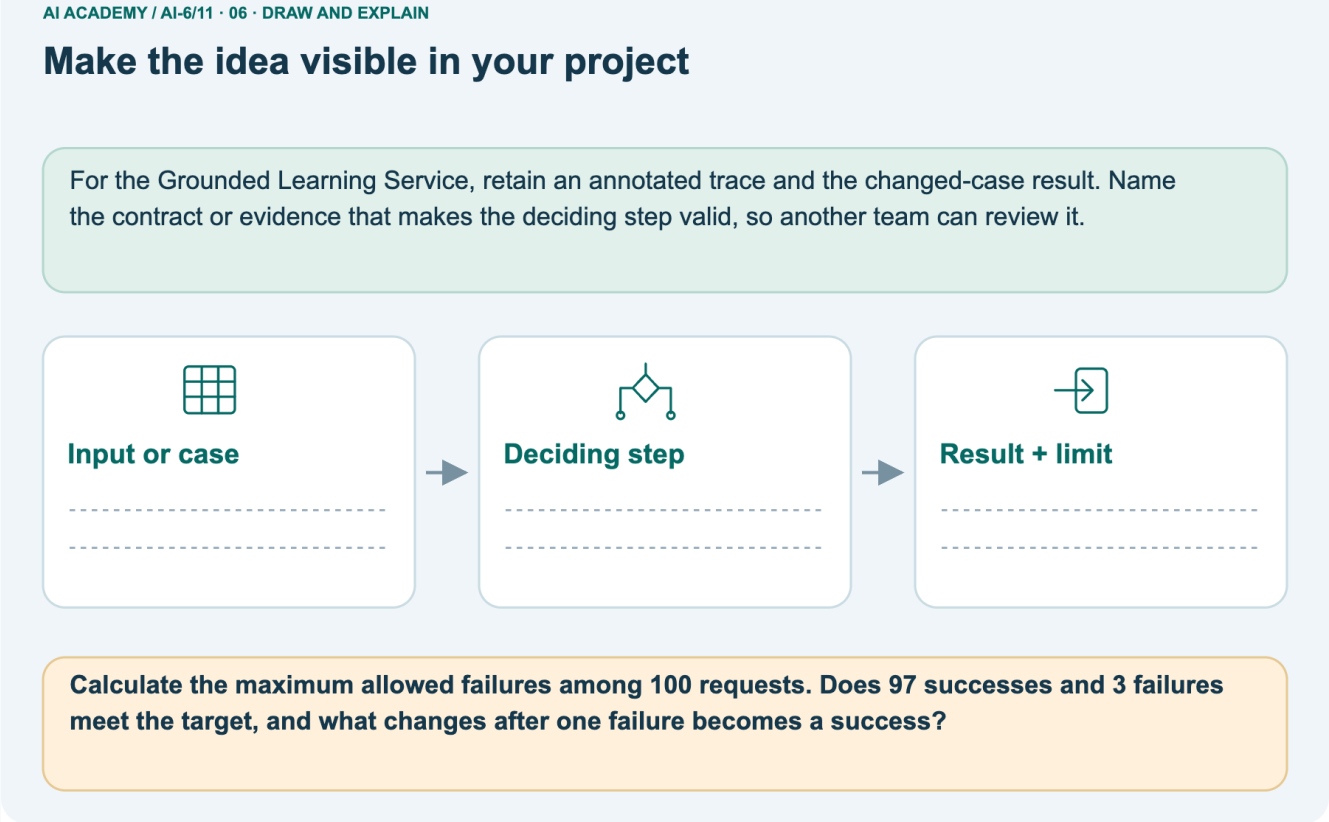
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Use the same predeclared correctness and timing criteria. The target permits at most 2 failures in 100 requests. Three failures misses the target by one request.

Changed case: 98 successes and 2 failures meet the stated window target. One recorded outcome changes while the denominator stays 100.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.